

Chapter 1 : Modules , Pip , Comments

What is a Module in Python?

A **module** is a file containing Python code (functions, classes, or variables) that you can **import** and use in other programs.

Modules help you:

- Reuse code.
- Keep your programs organized.
- Break down complex problems into smaller parts.

Types of Modules:

1. **Built-in Modules** – Already included with Python (no need to install).
 - Examples: math, random, datetime, os
2. **User-defined Modules** – Created by you (custom Python files).
 - Example: A file named calculator.py with your own functions.
3. **External Modules** – Created by others; must be installed using **PIP**.
 - Examples: numpy, pandas, requests, matplotlib

Code :

Example 1: Using a Built-in Module

```
print(math.sqrt(25)) # Output: 5.0
```

Example 2: Using Specific Function from a Module

```
from math import pow  
  
print(pow(2, 3)) # Output: 8.0
```

Example 3: Creating and Using Your Own Module

In file **my_module.py**:

```
def greet(name):  
  
    print("Hello", name)
```

In another file:

```
import my_module  
  
my_module.greet("Himanshu")
```

What is PIP?

PIP stands for “**Pip Installs Packages**”

It is the **Python package manager** used to install and manage **third-party modules**.

Install a package:

```
pip install package_name  
pip install NumPy  
pip install --upgrade package_name  
pip uninstall package_name
```

Comments:

Comments are used to explain or briefly describe instructions in our code.
Using comments makes our code more readable.

There are 2 types of comments:

- # : Single-line comments
- """ """: Multi-line comments

Practice Set

Que – 1 Import the random module and print a random number between 1 and 100.

Que - 2 Create a user-defined module calculator.py with an add (a, b) function.

Use it in another script.

Que – 3 Use pip to install the requests library (show the command).

Que – 4 Write a Python program that includes both single-line and multi-line comments.

Que – 5 From the math module, import factorial and calculate the factorial of 6.